Napier, 1 February 2019

How to Get Your Network Ready for 5G

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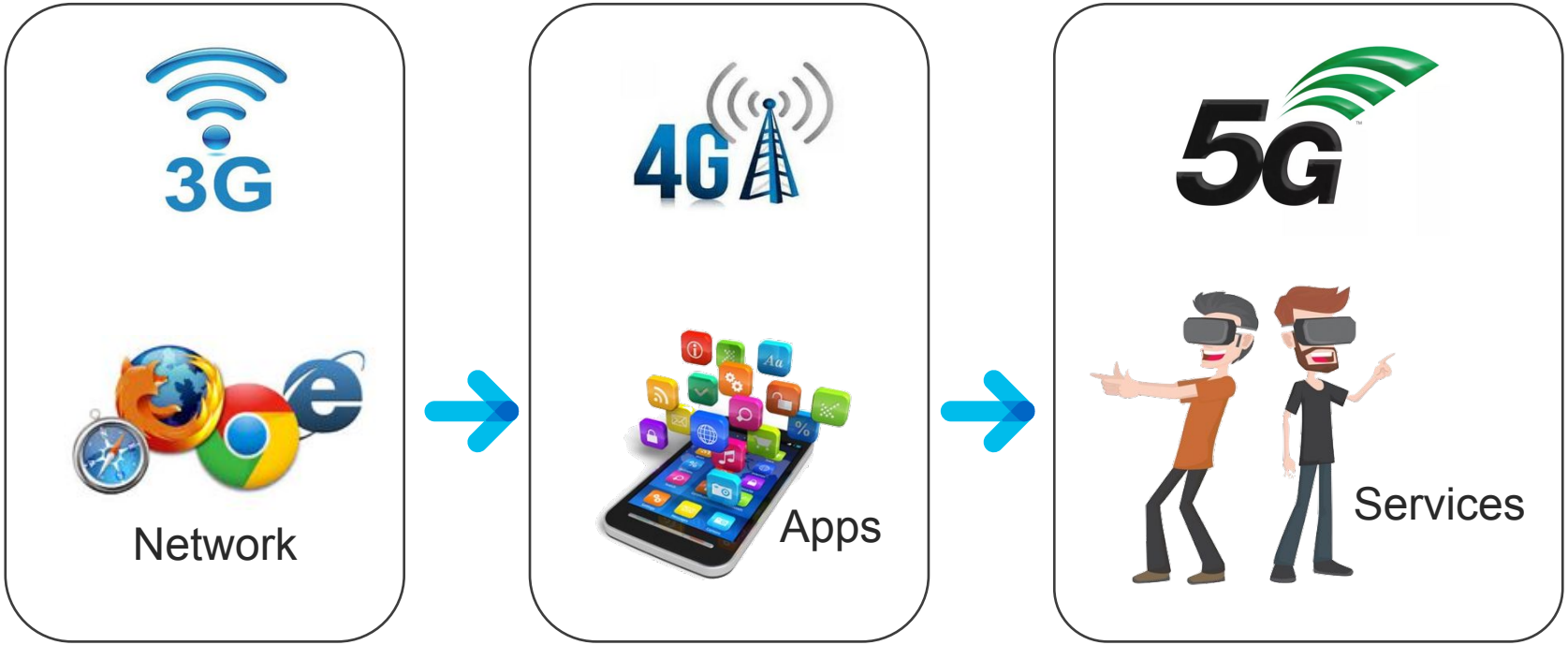
BEN HOWARD PHOTOGRAPHY
Are you ready
for the Fifth Frontier?

Mobile Service Transitions



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5G Service & Network Evolution

Enhanced Mobile Broadband
1000x BW, 10-100x
50 msec – 300 msec

Ultra Reliable Low Latency Communication
~1 – 25 msec Latency

Massive Machine Type Communications
1000x Density

Secure Remote Car Software Update

10 - 100M lines of code and hundreds of subsystems

Vehicle updates, telematics, and infotainment

Network-as-a-Service Public Sector Private Networks

Police, fire, hospitals with strict SLAs and Security

Smart City Enablement

Smart-sensor enabled

Waste and recycling,, parking, smart grid, homes

Augmented / Virtual Reality Delivery

Augmented, virtual, and mixed reality for learning, gaming, 4K/8K

Video enablement required

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Enablers for the Fifth Generation

Fast is good, but what good is getting more data if you are sloppy about what you do with it?,

Olivier Pauzet, VP/GM IoT Solutions, Sierra Wireless

Massive Bandwidth

ive Scale

Ultra-low Latency

Delivered Reliably

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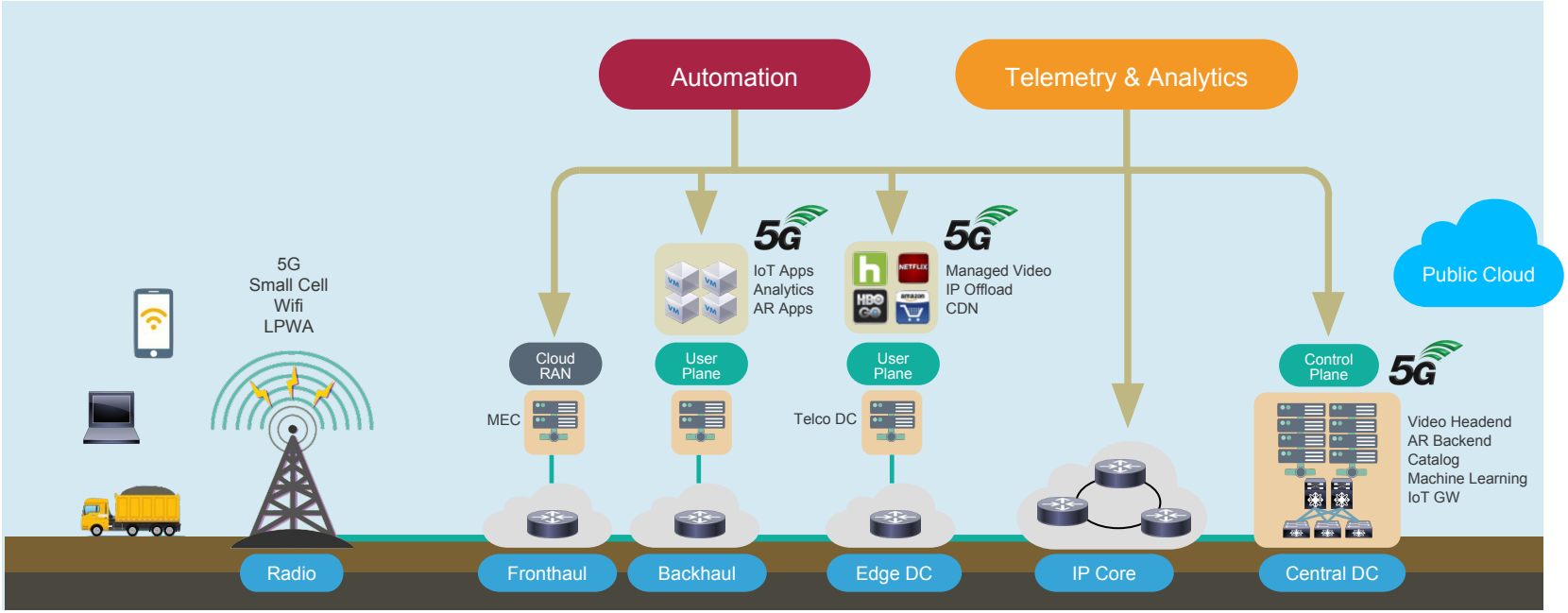
Evolved Landscape for 5G Services



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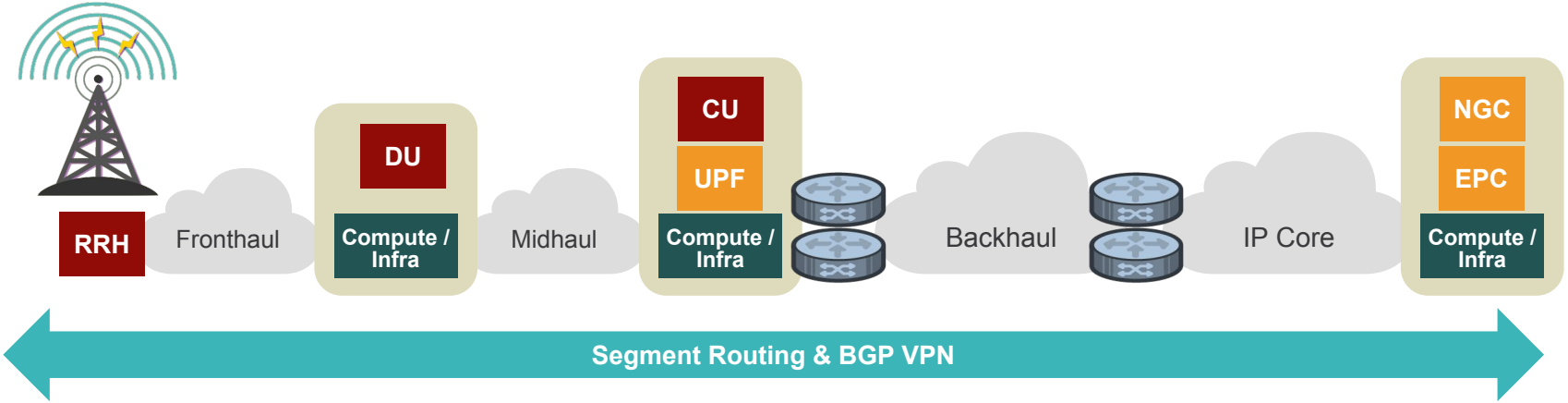
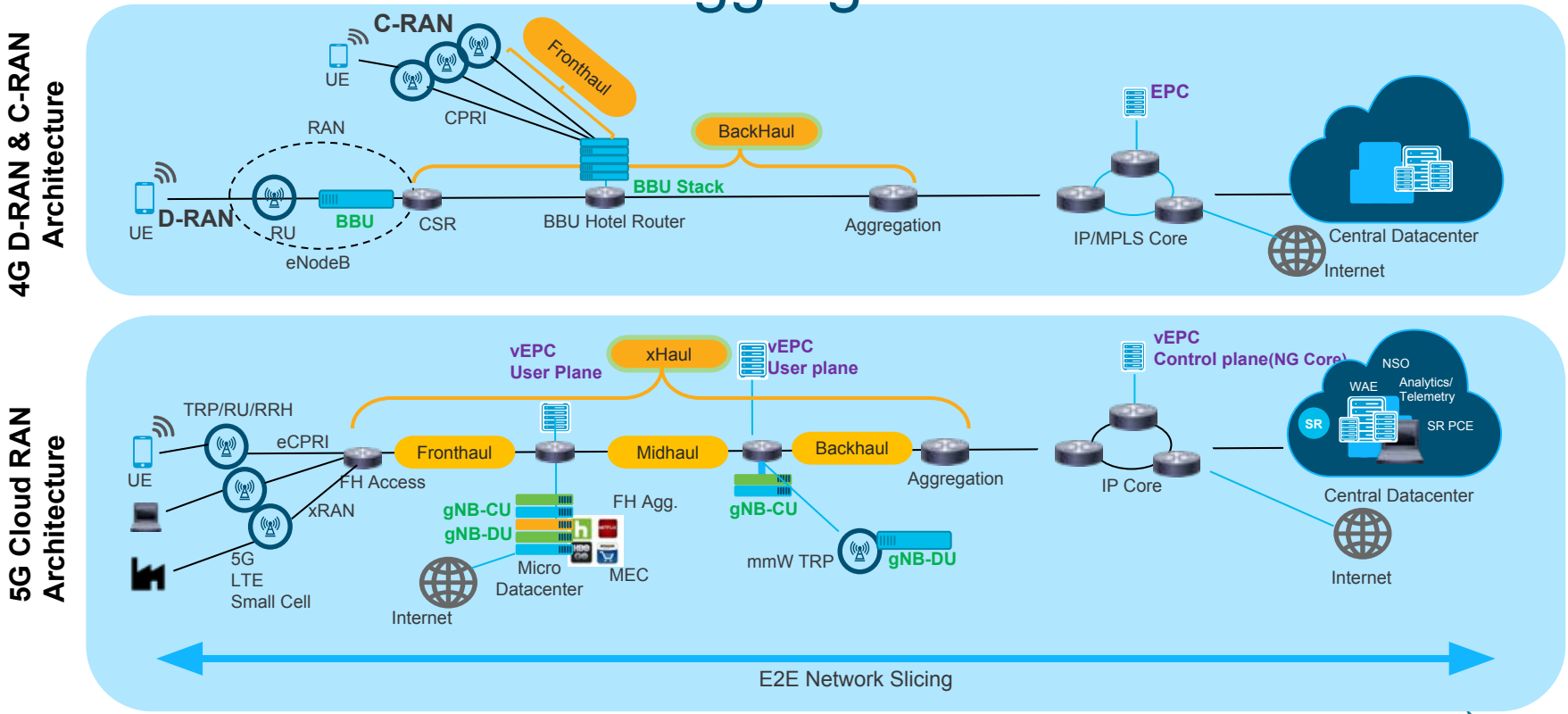
Key Capabilities of a 5G-ready Network

- Any-to-Any connectivity - Flexible service placement
- Efficient Service-aware Traffic Engineering
- Carrier Class Reliability including packet based clocking
- Reduced Network touch points
- Soft Network Slicing
- End-to-end Security



Mobile Evolution – Disaggregation & Virtualization



UE (User Equipment) **RU** (Radio Unit) RAN (Radio Access Network) **BBU** (Baseband Unit) **EPC** (Evolved Packet Core) **RU** (Radio Unit) **CSR** (Cell Site Router) **C-RAN** (Centralized RAN) **TRxP** (Transmit Receive Point aka Remote Radio Head (RRH)) **vEPC** (Virtual EPC) **CU-CP** (Centralized RAN Control Plane) **CU-UP** (Centralized Unit User Plane) **SR** (Segment Routing) **MEC** (Multi-access Edge Compute) **xHaul** (Backhaul + Midhaul + Sidehaul + Fronthaul) **FH Agg** (Fronthaul Aggregation Router) **FH Access** (Fronthaul Access Router) **D-RAN** (Distributed RAN) **mmW** (28GHz & 39 GHz) **Sub 6Hz** (600 MHz, 3.5GHz, 4.3 GHz)

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The Network Architecture is Evolving...

- Hierarchical
- North-South traffic
- Big Core groomed
- Centralized DC
- UMMT Architecture

- Flat / Any-to-Any
- East-West traffic
- More traffic stay in the Metro
- Distributed Content
- SR/SDN Architecture

Simplicity, Automation, Virtualization, Massively Scalable & Flexible

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Network Soft Slicing

Accommodating Different Apps on the Same Network

Massive Bandwidth

Massive Scale

Ultra-low Latency

Network Slicing

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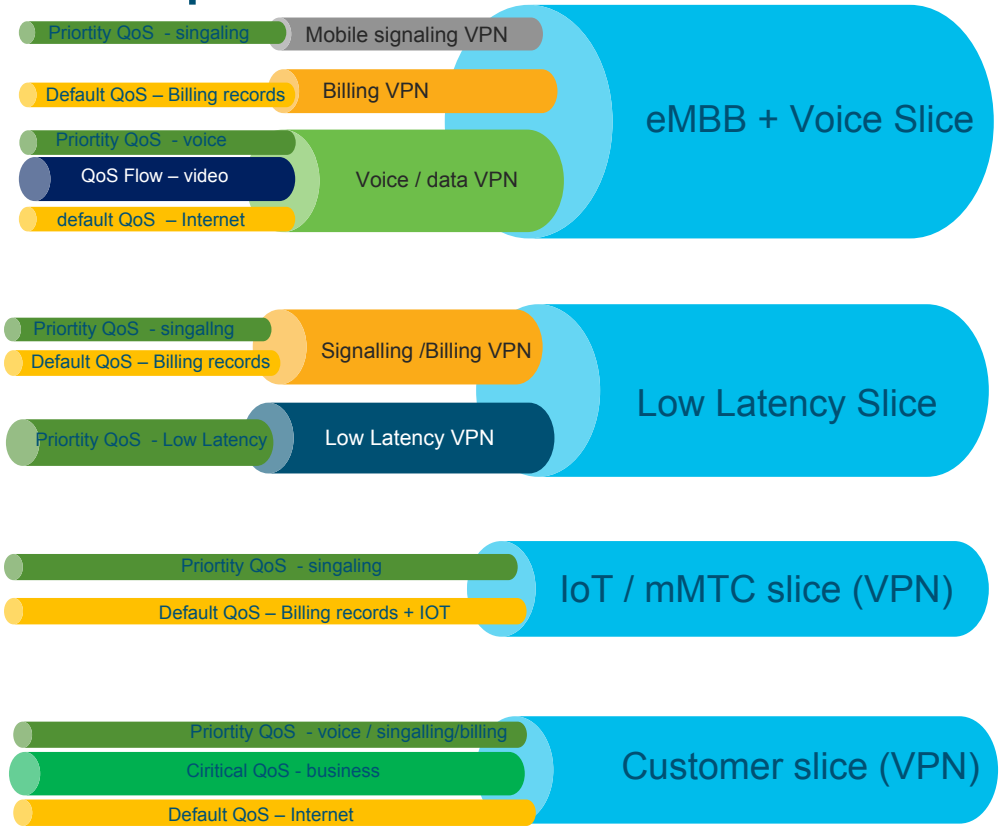
Example



Coverage
Multiple IP VPNs per slice
Multiple QoS classes within these VPNs:

Location specific
Dedicated to specific service
Low delay requirements

Coverage
Scaled for signalling capacity
Reduced need for mobility
Low bandwidths

Location specific
Dedicated to specific customer (MVNO/enterprise)

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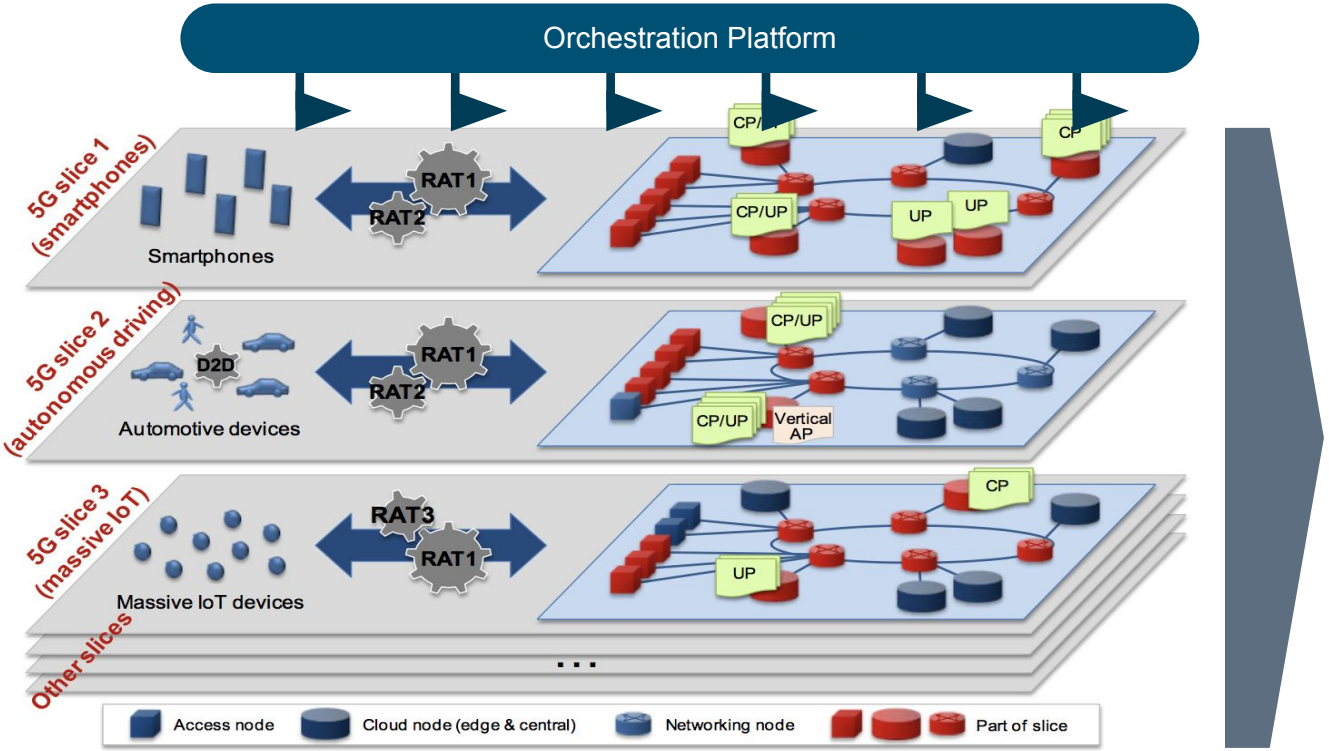
Network Slicing



Network Slicing is End-to-End

Source : NGMN 5G White Paper

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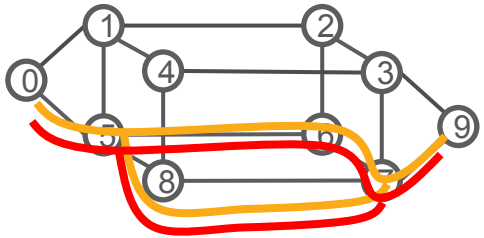
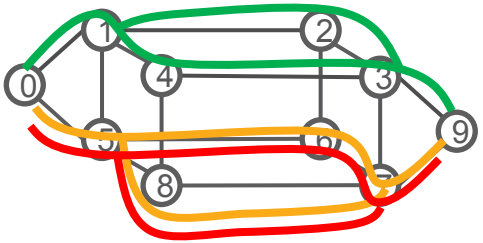
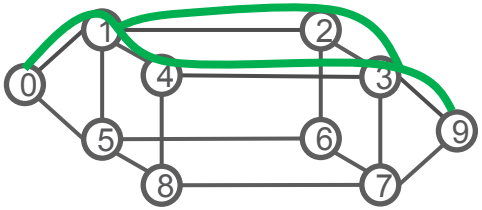
SR for Transport Network Slicing

Soft Slicing





Hard Slicing



- Hard slicing: Resources dedicated to a specific NSI. Data-plane resources allocated by a lambda, time domain multiplexed resource such as a FlexE channel or through a service such as an MPLS hard-pipe.
- Soft slicing: Provision of resources in such a way that whilst the slices are separated such that they cannot statically interfere with each other, they can interact dynamically which means they may compete for some particular resource at some specific time

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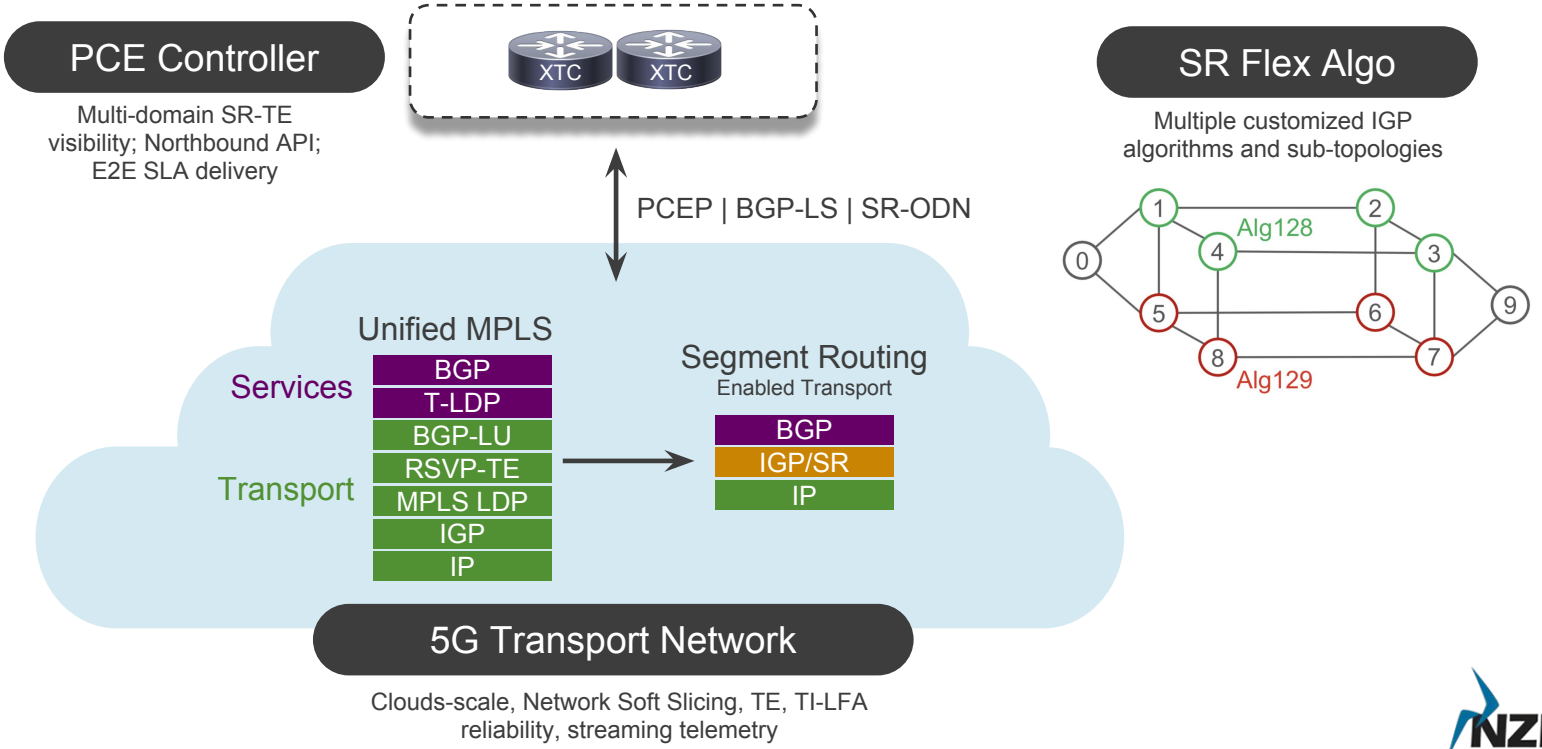
(see Network Slicing Architecture draft-geng-netslices-architecture-00)

SR-PCE 5G-ready Network Fabric



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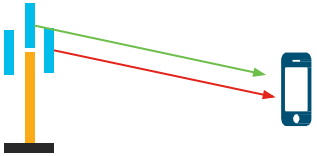
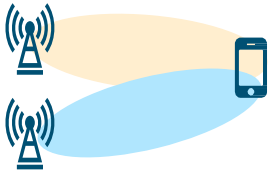
5G Synchronization Requirements



Transmission Diversity
±32.5ns Phase Accuracy
Improves error performance
Data Rate or Capacity



Carrier Aggregation
±65ns Phase Accuracy
Higher Pick Date Rate
More Bandwidth
Better Load Balancing

Coordinated Multi Point
±130ns Phase Accuracy
Higher Pick Date Rate
More Bandwidth
Better Load Balancing

Category	Type	Time alignment error (TAE) at antenna ports of eREs	TAE
A+	Relative	MIMO or TX diversity transmissions, at each carrier frequency	65 ns
A	Relative	Intra-band contiguous carrier aggregation, with or without MIMO or TX diversity	130 ns
B	Relative	Intra-band non-contiguous carrier aggregation, with or without MIMO or TX diversity, and Inter-band carrier aggregation, with or without MIMO or TX diversity	260 ns
C	Absolute	3GPP LTE TDD	3 us

cTE	Accuracy Budget
Class A (existing)	50ns
Class B (existing)	20ns
Class C (proposed)	10ns

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eRE (eCPRI Radio Equipment) **cTE** (Constant Time Error) **MIMO** (Multiple Input Multiple Output)

5G Creates Scale & Change

	Stone Age	Last Century	Last Decade	This Decade	Today	Tomorrow
Technology Trends	**Mainframe**	**Client Server**	**Distributed**	**Virtualization**	**Cloud**	**Digital Business**
Device Counts	1	10s	100s	1,000s	10,000s	100,000s
Deployments/Year	1	2	10s	100s	1,000s	10,000s
Monitoring Tools	1	3	5	10	25	50+
Events/Metrics/Day	**100s**	**1000s**	**100,000s**	**Millions**	**Billions**	**Trillions**
Organizational Silos	1	10	15	25	50	100
Humans Ability to Cope	**Yep**	**Yep**	**Kind Of**	**Not Really**	**Nope**	**HELP!**
Service Availability	100%	99.999%	99.99%	99.9%	99%	?

Where are you today?

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From the Old World:

Physical devices described by rules, mapped out in brittle static representations

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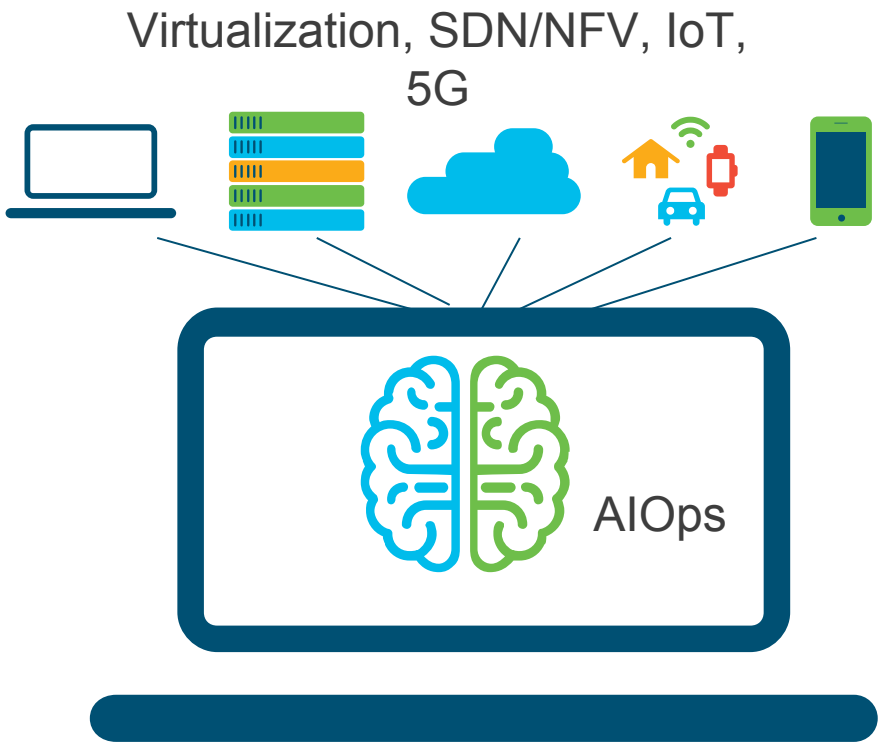
To the New World:

Dynamic algorithms replace rules to manage ever-increasing rate of change



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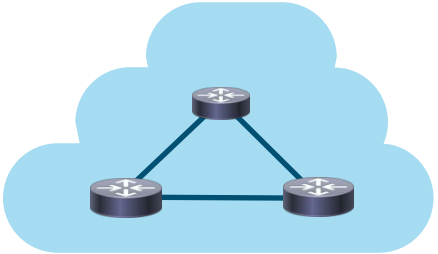
Freeing the Data with Streaming Telemetry

Instruction on:
- What data to collect
- With what cadence
- And send to where

Network Infrastructure



Push

Streaming Telemetry

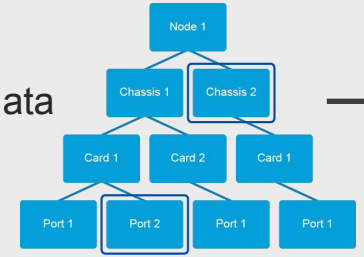
As much Data
As Fast
As Useful
As Easy
As Possible

Analytics-Ready Data

Machine Learning
Deep Learning
AI Ops

Model-Driven Telemetry

YANG Data Models:
- Explicitly and precisely defines data
- Consistent and complete




```
list interface {
    key "interface-name";
    leaf interface-name {
        type string;
    }
    leaf speed {
        type string;
    }
    leaf duplex {
        type string;
    }
}
```

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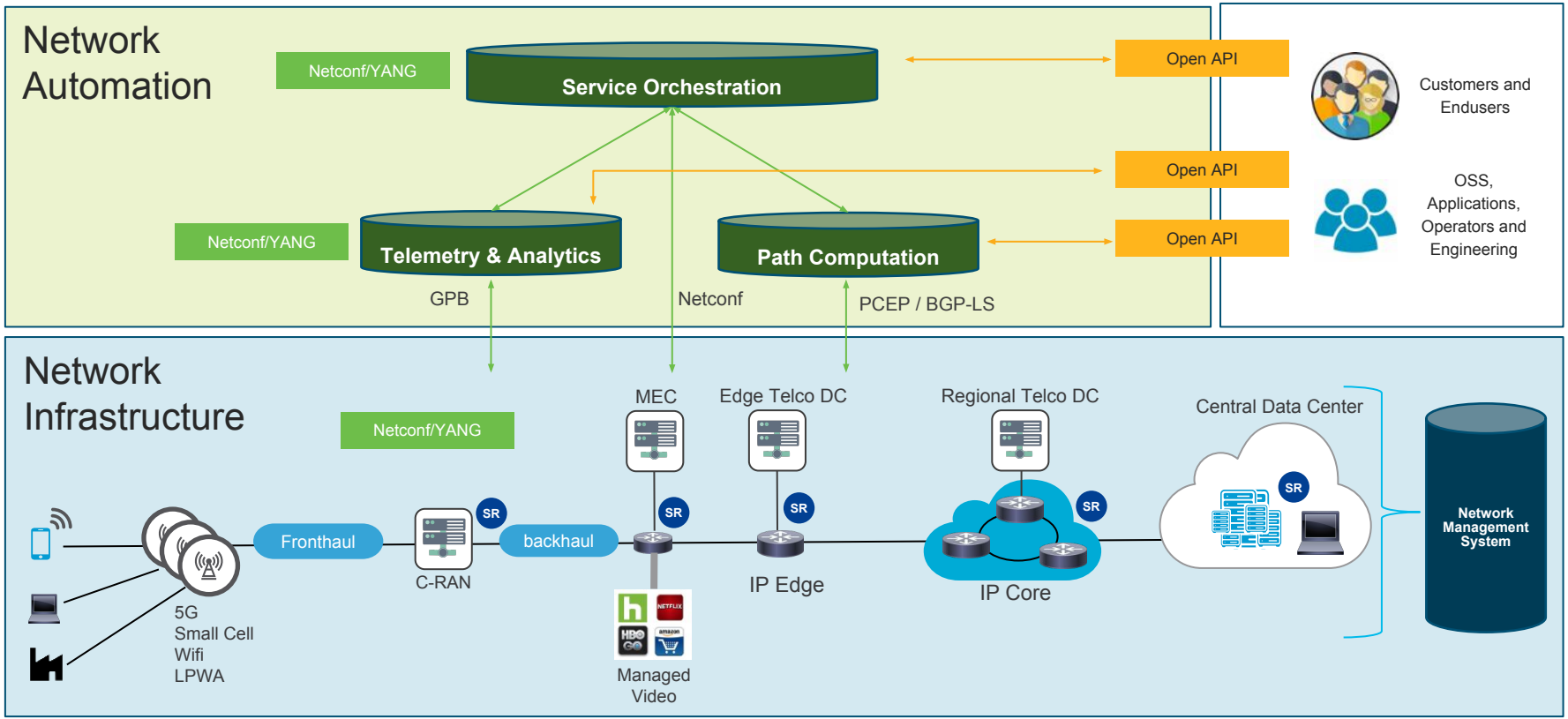
Network-as-a-Service for 5G Services

From CLI to Netconf/YANG and Network API's



Summary

- 5G expands Telecommunications beyond Consumer Mobility as we know it today. It will herald deep **Network Architecture Evolution.**
- **End-to-end Thinking** becomes Critical.
- New **Programmable Networks** with **NetOps/AIOps** approaches to handle scale, dynamism.

Thank you. Enjoy NZNOG. :-)

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